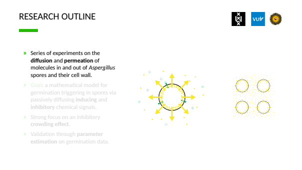
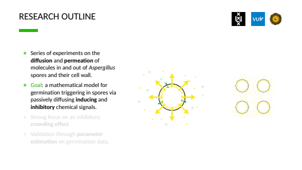
2025-07-24

Diffusive Signalling for *Aspergillus* Germination

1. Welcome everyone and thank you for coming to the defence of my thesis "Modelling Diffusive Signals for the Germination of *Aspergillus* conidia", a topic which binds together Biology, Computation and Molecular Physics, among other fields. I will begin with a research outline and an introduction to the topic of *Aspergillus* moulds and the germination of their asexual spores, the conidia. This will be followed by a brief overview of the computational methods used to simulate diffusive germination signals. Then I will present all the computational experiments performed, together with the insights obtained from them, leading to valuable conclusions regarding relevant questions from the research field.



1. The thesis does not target a single hypothesis but rather a range of research questions that deal with the topic of diffusion and its derivative process, barrier permeation, in the context of conidial germination.
2. The ultimate goal is to construct a mathematical model that describes how passively diffusing inducing and inhibitory signals trigger spore germination.
3. A special focus of this research is the inhibitory crowding effect in spore cultures, on which I will elaborate later.
4. As a final step, the constructed germination models are fitted on experimental data, aiming to test their validity and to shed light on some likely scenarios behind germination.



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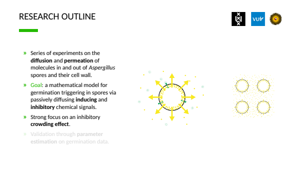

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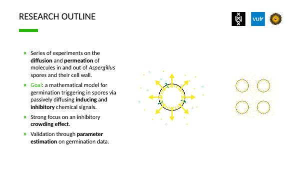

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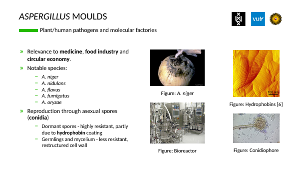
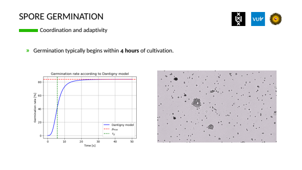
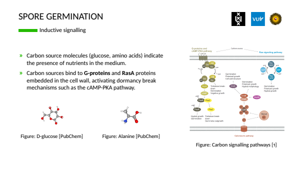


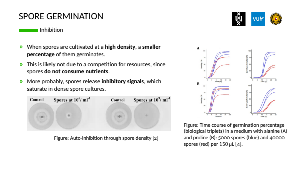
1. The *Aspergillus* moulds are ubiquitous in our environment. In fact, they are so common, that you are likely breathing in some of their spores as we speak. You have likely seen one of their representative species, *A. niger*, as the black mould forming between onion layers. That same mould is used in industrial bioreactors for the production of citric acid and other useful substances. This is done at such scales that most of the citric acid in the world is produced by this mould rather than from citric fruits. Other representatives are used in fermentation processes, such as *A. oryzae*, for the production of soy sauce. Very often, however, *Aspergillus* moulds can be pathogenic to plants and humans, making them a double-edged sword for the food industry and for medicine.
2. The primary characteristic of *Aspergillus* moulds are their asexual spores, the conidia, which can survive in extreme environments for a long time while dormant, most notably due to their outer hydrophobin coating. Once the spores germinate, they turn into vegetative mycelium, which can colonise larger areas but is more vulnerable.



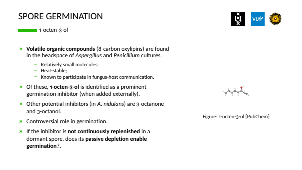
1. When inoculating a favourable environment, spores typically germinate within a few hours, but due to their largely heterogeneous physiology, the germination times vary. Some established models, like the Dantigny model, capture the percentage of germinated spores over time as a sigmoidal curve with a characteristic onset time and a maximum germination percentage.
2. During germination, the conidia swell, take up water and form polar protrusions called germ tubes. The germ tubes then gradually form the mycelium of the mature colony.



1. One of the signals responsible for germination is linked to the presence of potential nutrients - degradable molecules containing carbon or nitrogen, for example glucose or amino acids. It is presumed that such molecules bind to receptor proteins embedded inside the conidial cell wall and trigger complex pathways that lead to the transformation of the spore.



1. Experiments have shown that when a large number of spores are cultivated in a limited space, a smaller percentage of them manage to germinate. Since dormant spores do not consume nutrients, this cannot be due to competition for resources, and it is therefore suspected that they release inhibitory molecular signals which saturate in the presence of many spores. This CROWDING EFFECT of reduced germination is thought to be a BET HEDGING mechanism to maintain spore reserves in case of unfavourable environmental changes.



1. A primary suspect for this inhibition is 1-octen-3-ol, a volatile organic compound which is commonly produced by fungi and is partly responsible for the characteristic smell of mushrooms. The size of its molecule is similar to glucose, which makes it easily diffusible in water. There are several other very similar volatile organic compounds found in the headspace of *Penicillium* and *Aspergillus* cultures, but 1-octen-3-ol has been shown to have the strongest inhibitory effect.
2. Nonetheless, its exact role is disputed up to this date. Researchers have shown that if it is added to a spore culture at a given dose, it can prevent germination. This dose, however, seems to be much higher than what can be extracted from a spore culture. So the question is, if a dormant spore contains a non-replenishable amount of 1-octen-3-ol, does its passive depletion enable germination?

RESEARCH QUESTIONS

Are signals timed by diffusion/permeation?

» The flux J of molecules through a homogeneous medium is characterised by a **diffusion constant** D:

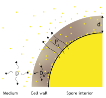
$$J = -D\frac{\partial c}{\partial x}. \tag{1}$$

» The passage of molecules through a barrier is determined by a **permeation constant** $P_s \sim D/d$:

$$J = -P_s \Delta c. \tag{2}$$

» In both cases, the difference in the **concentration** c is driving the flux.

» Do **known diffusivities/permeabilities/concentrations** lead to signal **timings consistent with germination**?



1. This leads to the fundamental question, if the cell wall or the extracellular medium has limiting diffusivity, how long does it take for the inhibitor molecules to leave the spore? And vice versa, do inducing signals reach the receptors immediately, or are they also slowed down by diffusion?
2. The mathematics of diffusion is a long-studied subject, essential for modelling biophysical processes. Using the language of diffusion, we can use two constants to calculate the time for redistribution of molecules: the diffusion constant D, which indicates how the mean squared deviation of a particle grows with time, and has units of area per time; and the permeation constant P_s, which is applicable to semi-permeable barriers, relates D to a specific barrier thickness and has units of speed. Furthermore, the difference or gradient of concentrations in space is a driving component for diffusion, as any molecular species wants to maximize its entropy by dispersing as evenly as possible.
3. Thus, using some known values for D, P_s and the initial and threshold concentrations of substances, we can see if diffusive signalling makes sense in the context of germination.

RESEARCH QUESTIONS

Is diffusion/permeation modulated by (de)saturation?

- Inhibitor molecules diffuse as time-dependent Gaussian distributions around the sources.
- A denser array of sources causes inhibitor **saturation** in the extracellular medium.
- When the concentration in the medium is **similar** to the concentration in the spore, the molecules diffuse **slowly**.
- Can **increased spore densities** change germination behaviour via external **inhibitor saturation**?

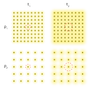
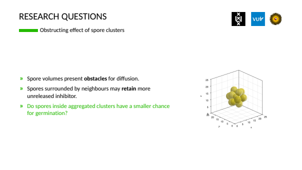
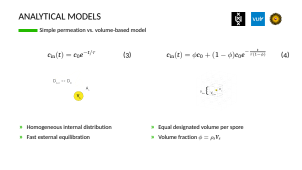


Figure: Saturation at different times and spore densities.

1. When spores release inhibitor, they also gradually fill the external medium, until the internal and external concentrations equilibrate. It makes sense that the more spores release inhibitor, the higher the equilibrium levels become, and so conidia may stop releasing inhibitor molecules before they have reached the depletion threshold necessary for germination. The question is, at what spore densities does this happen?



1. Furthermore, spores are also spatial obstacles for diffusion. It is known that the conidia of e.g. *A. niger* agglomerate into clusters in the early stages of cultivation. Is it then reasonable to believe that the spores deep inside these clusters cannot release their inhibitor as efficiently as solitary spores?



1. To tackle these questions, a framework of computational models was constructed, combining analytical formulas with lattice-based numerical simulations to compute different scenarios.
2. First, a simple permeation model is applied, widely used in pharmaceutics and biophysics, which views the spore as a "balloon" full of well-mixed substance, with a semi-permeable boundary which releases molecules much slower than their dispersal in the outside medium. Therefore, the outside concentration is assumed to be constant, at zero. This leads to an exponential decay of the original concentration, with a characteristic time τ proportional to the volume, surface area and permeation constant of the barrier. This formula is used for the inhibitor permeation, but a modification can be applied for the inducer permeation.
3. While this model can easily compute the concentration evolution in a single spore, it fails to capture the phenomenon of external saturation in dense spore cultures. An extension of it discards the assumption that concentrations diffuse to zero in the infinite outside medium, but rather assigns an equal designated volume to each spore in the culture, which varies depending on the density of spores. This extension is dubbed *the volume-based model*.

NUMERICAL MODELS

Model resolutions

» Spore representation on macroscopic, spore and cell-wall scale

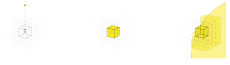
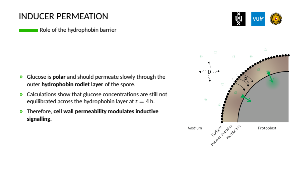


Figure: Low resolution.

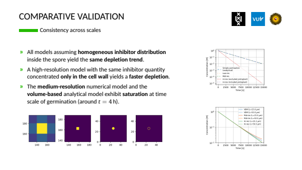
Figure: Medium resolution.

Figure: High resolution.

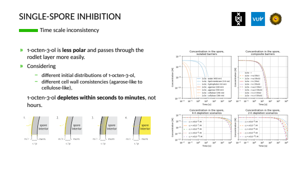
1. On the other hand, diffusion simulations are an intuitive way to trace the evolution of concentrations over time. Three numerical models are constructed to represent diffusion on three different scales - macroscopic (low resolution), spore-scale (medium resolution) and cell-wall scale (high resolution).



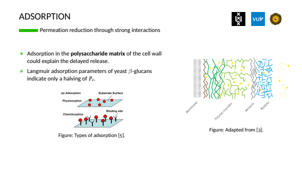
1. The simplest test case uses the permeability of the hydrophobin rodlet layer to glucose molecules to show that the inducing signals become significantly slowed down on their way to the receptors. At 4 hours, a typical glucose concentration still has not fully equilibrated across the outer layer.



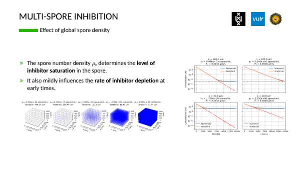
1. An important step in validating all models is to run them with equivalent parameters and see if they produce consistent results. All of them reproduce the same exponential decay of the inhibitor concentration, but, additionally, the high-resolution model allows a more nuanced view of the inhibitor distribution. For example, if the inhibitor is only confined within the cell wall, as opposed to being homogeneously distributed inside the spore, the release happens much faster due to the altered concentration "pressure".
2. Furthermore, the medium-resolution and the volume-based model reproduce the effect of inhibitor saturation in dense spore cultures, as can be seen in the tails of these graphs.



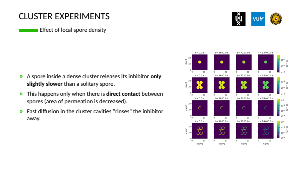
1. On the other hand, 1-octen-3-ol is much less polar than glucose and supposedly passes through the main barrier of the hydrophobin layer much faster. Regardless of how it is distributed in the spore or the cell wall, or what assumptions are made for the rest of the cell wall, it appears that 1-octen-3-ol dissipates too quickly away from the spore - within seconds to minutes, instead of hours.



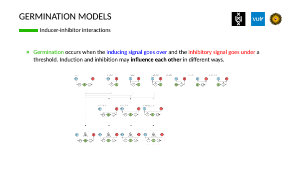
1. This leads to the suspicion that something else slows down the passage of 1-octen-3-ol on its way out of the spore. One possibility is that the polysaccharide matrix of the cell wall binds strongly to the inhibitor molecules, leading to adsorption effects. Using some known adsorption parameters of yeast β-glucans, a possible halving of the effective permeation constant is possible, which is still insufficient to bring the depletion to the germination time scale.
2. Therefore, the adsorption in *Aspergillus* spores is either much stronger, or another limiting phenomenon needs to be accounted for - the permeability of melanin, a significant time lag between depletion and germination, or perhaps a continuous synthesis inside the dormant spore.



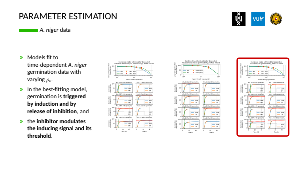
1. In any case, if the permeability of the cell wall is assumed such that the inhibitor is depleted within 4 hours in a single spore, then in dense cultures it is shown that the number density of spores determines the saturation level of residual inhibitor in the spore (green lines in the right plots). So the depletion trend starts as exponential in the early times but then saturates to an equilibrium value at late times.



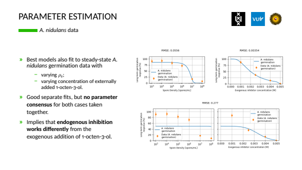
1. This effect only appears when the spore density changes globally — in isolated spore clusters, the inhibitor is quickly rinsed through the intermediate cavities, and only a faint difference in inhibitor retainment is noticeable, mostly due to the permeation area that is sealed by neighbours in direct contact.



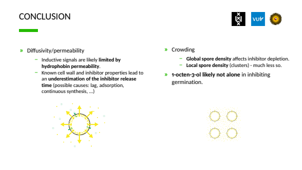
1. Having established the mechanisms of diffusion and the factors relevant for germination, different models are proposed which combine induction and inhibition as binary triggers for germination. In the simplest case, germination occurs when both the inducing signal surpasses a threshold AND the inhibitory signal goes under a threshold. However, further options are explored in which the inducing signal (here, blue dots) and the inhibitory signal (here, red dots) modify each other’s strength or threshold.



1. All these germination models are tested in their ability to represent actual time-dependent germination data with *A. niger* spores, where the density of the spore culture is varied. This is done through global and local optimisation algorithms. One criterion for success is the RMSE value of the fit and another one is the sensibility of the resulting parameters. Taking both of these into account, it is found that the most valid model is one where the inhibitor modulates the inducing signal and its threshold, but also induction and the release of inhibition directly trigger germination.



1. Something interesting happens when we try to fit these same models to a different data set, one with *A. nidulans* spores, where germination is controlled on one hand through varying culture densities, and on another hand by simply adding different concentrations of 1-octen-3-ol to the culture. It then turns out that the model can be successfully fit to either the one case or the other, but not to both simultaneously. This is a clear indication that the inhibition caused by artificially adding 1-octen-3-ol is different from the mechanism of actual auto-inhibition.



1. To sum up, the research presented so far leads to several interesting findings. First of all, the hydrophobin coating of spores is impermeable enough to limit the access of carbon sources to the receptor proteins inside the cell wall, but not impermeable enough for 1-octen-3-ol to slowly escape the spore at the time scale of the germination onset.
2. Second, the global spore density is shown to affect the concentration gradient that drives the inhibitor release. But the local density, in terms of dense spore clusters, does not have a strong effect, unless the contact area between spores is increased.
3. Most notably, some of the proposed germination models are capable of explaining germination data under sensible parameters, but the assumption that 1-octen-3-ol acts alone in inhibition is wrong, since otherwise the model would otherwise equivalently represent inhibition through exogenous addition of the compound.
4. Naturally, these are all in silico experiments using data sourced from a various publications, but hopefully they provide a stepping stone in compiling knowledge about diffusion and germination, and highlight some interesting questions to be pursued further.

1. With this, I would like to conclude my presentation. I would like to thank all of you for attending, special thanks to my supervisor Han Wösten for the guidance in this complex topic, and to everyone else who supported me during the last year. This is only a brief summary of the entire research, so I would be happy to take your questions if you want to know more.